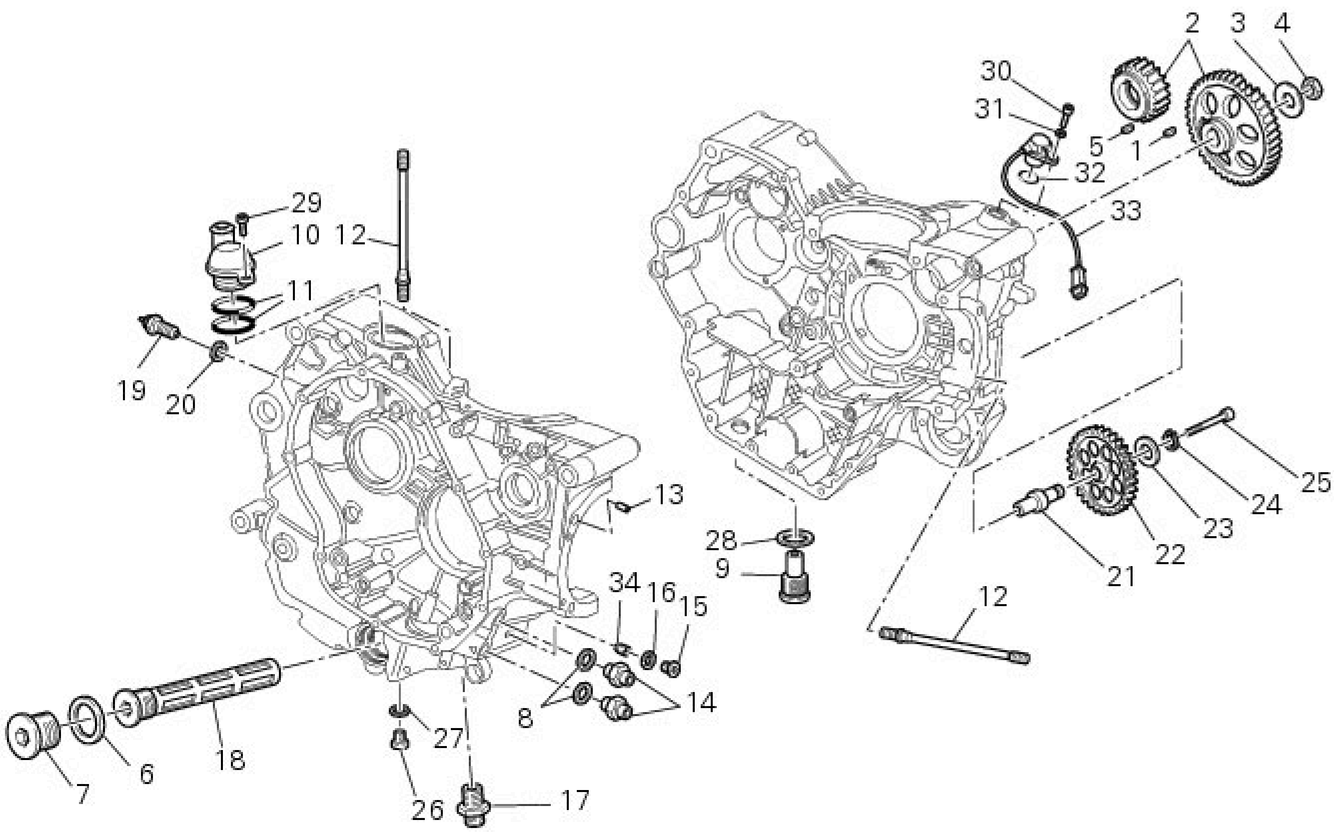
9.1 -Crankcase assembly: external components



1 Key
2 Timing gear pair
3 Lock washer
4 Hexagonal nut
5 Key
6 Sealing washer
7 Plug
8 Aluminium gasket
9 Plug
10 Breather valve
11 O-ring
12 Cylinder head stud
13 Locating dowel
14 Nipple
15 Plug
16 Sealing washer, thickness 2
17 Nipple
18 Mesh filter
19 Neutral switch
20 Sealing washer
21 Gear shaft
22 Idle gear
23 Washer
24 Circlip
25 Screw
26 Plug

27 Sealing washer
28 Sealing washer
29 Screw
30 Screw
31 Washer
32 O-ring
33 Pick-up sensor
34 Grub screw

Spare parts catalogue

1100 ABS	TIMING SYSTEM
1100 ABS	FILTERS AND OIL PUMP
1100 ABS	CRANKCASE HALVES
1100 ABS	ELECTRIC STARTING AND IGNITION
1100 ABS	GEARCHANGE CONTROL
1100 S ABS	TIMING SYSTEM
1100 S ABS	FILTERS AND OIL PUMP
1100 S ABS	CRANKCASE HALVES
1100 S ABS	ELECTRIC STARTING AND IGNITION
1100 S ABS	GEARCHANGE CONTROL

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

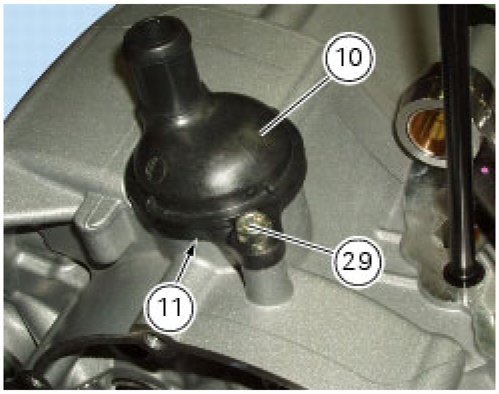
Removing outer components

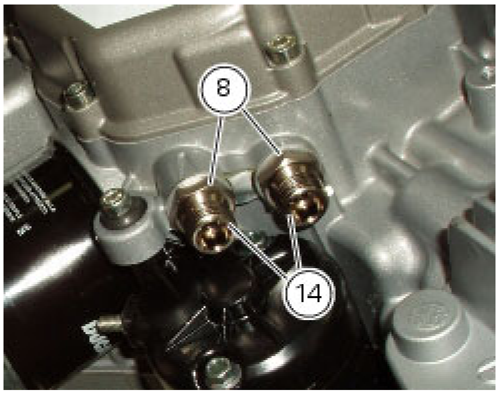
Operation	Section reference
Remove the engine from the frame	N 1, Removal of the engine
Remove the lubrication system	N 2.1, Removal of the oil pump
Remove the timing belts and the timing belt pulleys	N 4.2, Removal of the timing belt covers
Remove the camshafts	N 4.3, Removal of the camshafts
Remove the complete cylinder head unit	N 4.4, Removal of the cylinder head assembly
Remove the cylinder barrel/piston assemblies	N 5, Removal of the cylinder/piston assembly
Remove the alternator-side crankcase cover and the alternator assembly	N 8, Removal of the generator cover
Remove the starter motor	P 3, Removal of the starter motor

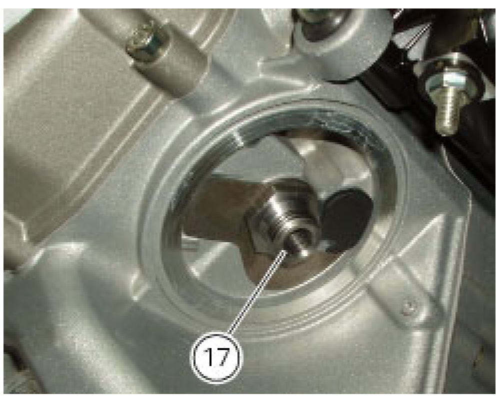
Note

The following removal operations are required in order to renew and/or clean the crankcase halves. If the original crankcase halves are to be reused, then the removal of these components is not essential.

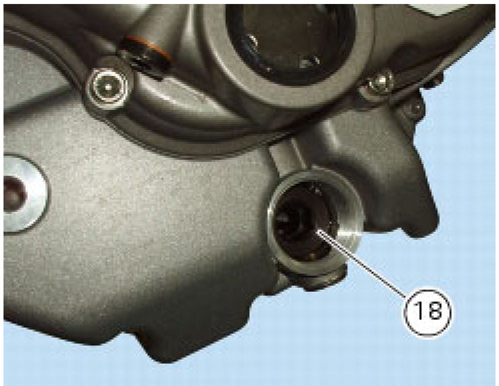
Unscrew the screw (29) and remove the oil breather valve (10) with the O-rings (11). Check the condition of O-rings (11) and renew them if necessary.



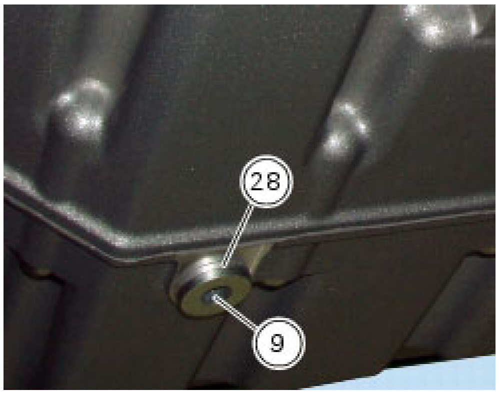
Unscrew and remove the oil inlet and outlet nipples (14) from the clutch-side crankcase half and recover the seals (8).


Unscrew and remove the oil filter cartridge (Sect. D 4, Changing the engine oil and filter cartridge).
Unscrew and remove the oil filter support nipple (17).


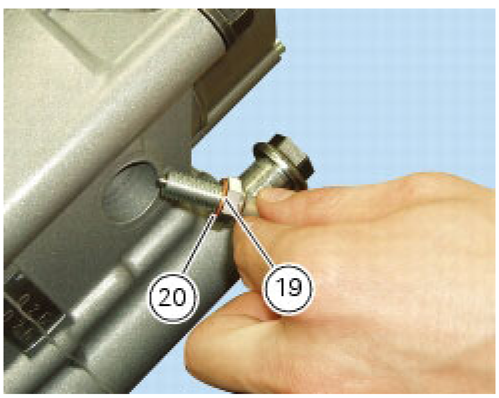
Remove the mesh filter (18) with its seal from the clutch-side crankcase half as described in Section D 4, Changing the engine oil and filter cartridge.
Loosen plug (26) taking care to recover the washer (27).

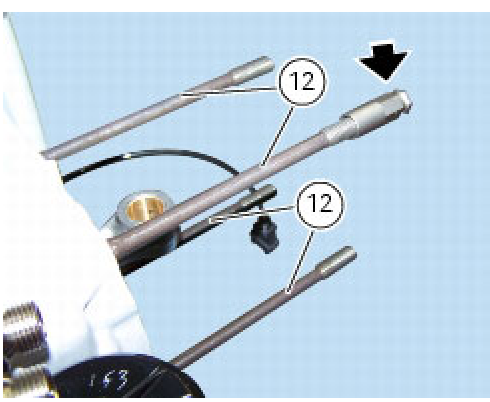
Remove the drain plug (9) with its seal (28).



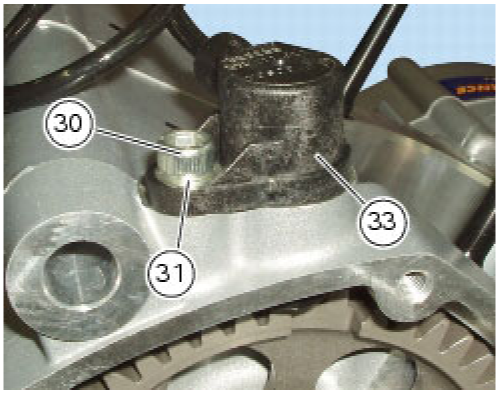
Remove the neutral switch (19) with seal (20).



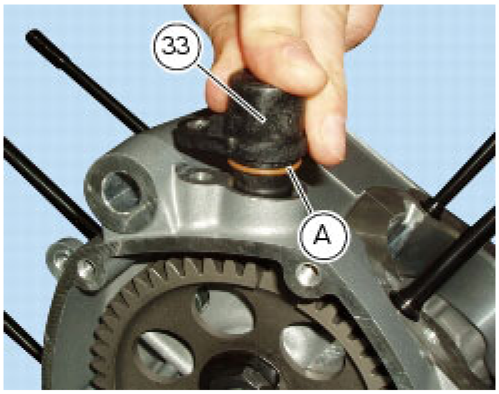
Remove the cylinder head studs (12) with the aid of the appropriate tool.

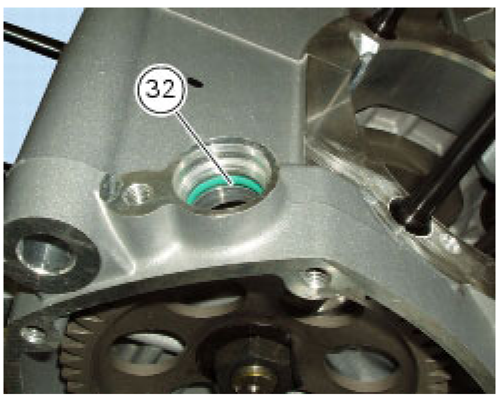


To remove the engine sensor (33), unscrew the screw (30) and recover the washer (31).



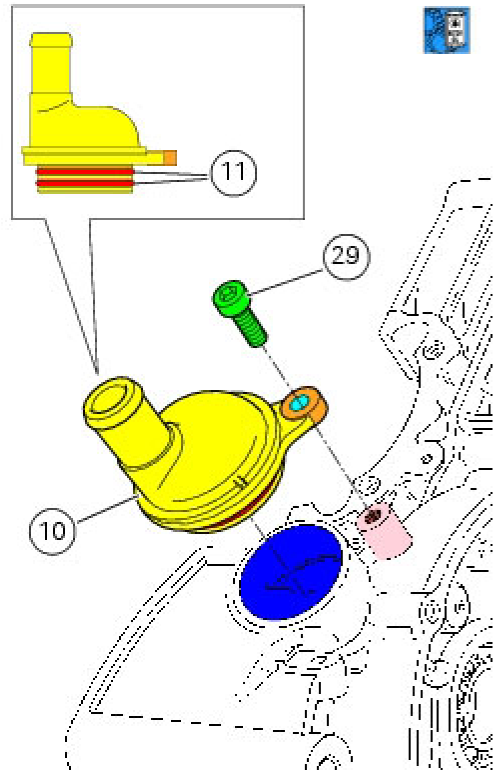
Check the condition of the O-ring (32) in the crankcase half and the O-ring (A) on the sensor (33) and renew them if necessary.

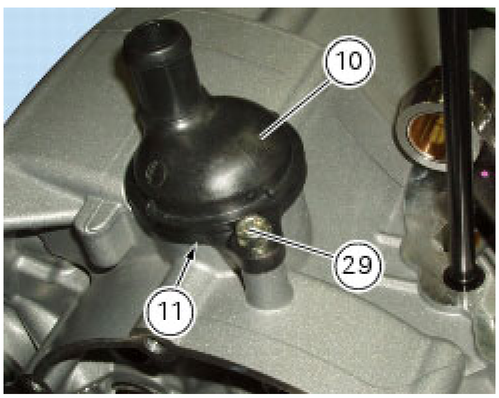




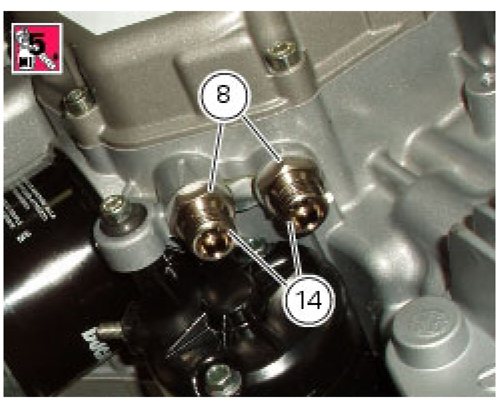
Refitting the external components

Check the condition of O-rings (11) and renew if necessary.
Install the oil vapour breather valve (10) in the crankcase along with O-rings (11), previously lubricated.
Tighten the screw (29) to the specified torque (Sect. C 3, [Engine torque settings]).

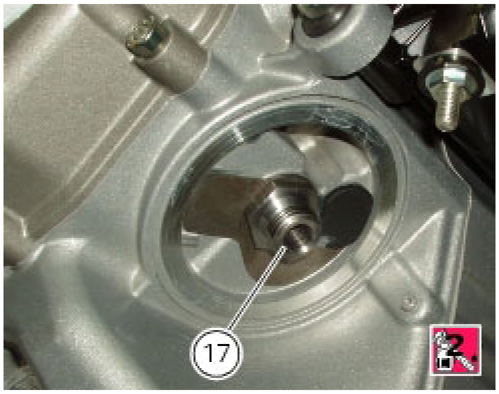




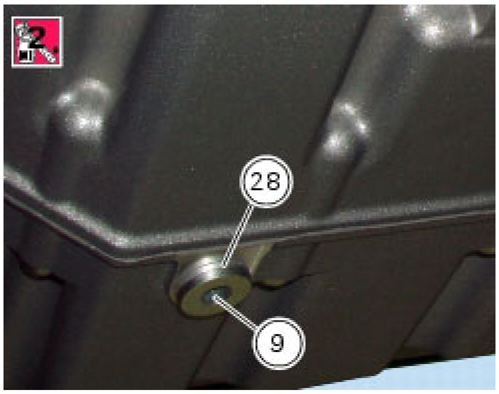
The sealing washers (8) must be installed on nipples (14), on the side of the nipple with the smaller diameter thread.
Apply the recommended threadlocker to the smaller diameter thread of nipples (14).
Tighten the nipples (14) to the specified torque (Sect. C 3, Engine torque settings).



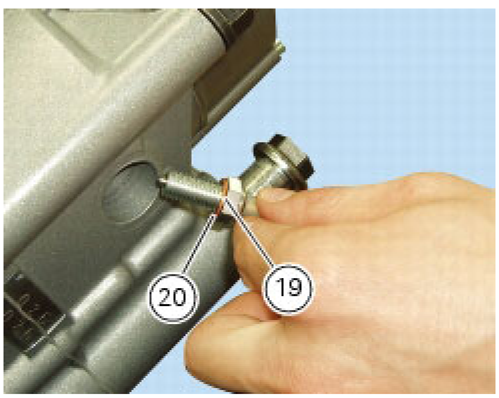
Apply the recommended threadlocker to the nipple (17).
Tighten the oil filter cartridge support nipple (17) to the specified torque (Sect. C 3, Engine torque settings).



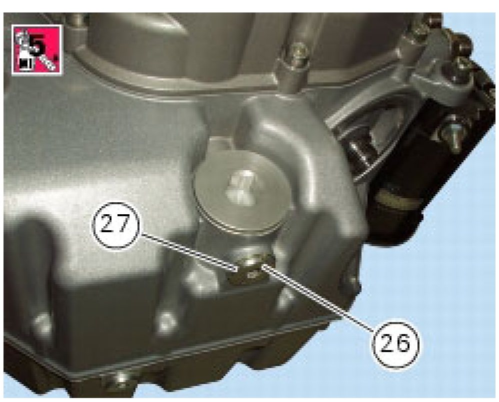
Fit the drain plug (9) with its seal (28) and tighten to the specified torque (Sect. C 3, Engine torque settings) applying the specified threadlocker.



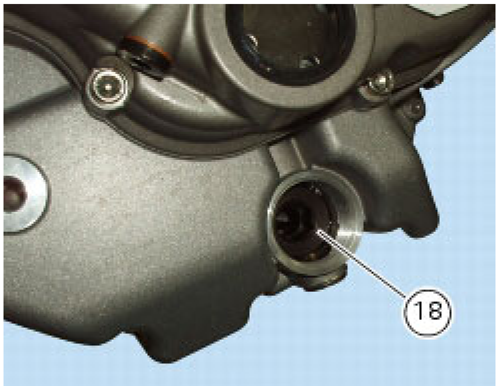
Refit the neutral switch (19), fit the seal (20) and tighten to the specified torque (Sect. C 3, Engine torque settings).



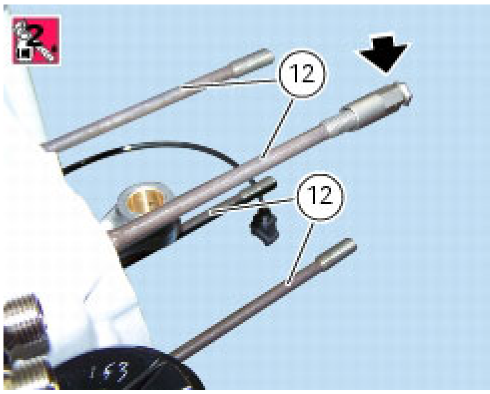
Apply prescribed threadlocker on the plug (26), insert it in place with its gasket (27) and tighten the plug to the specified torque (Sect. C 3, Engine torque settings).



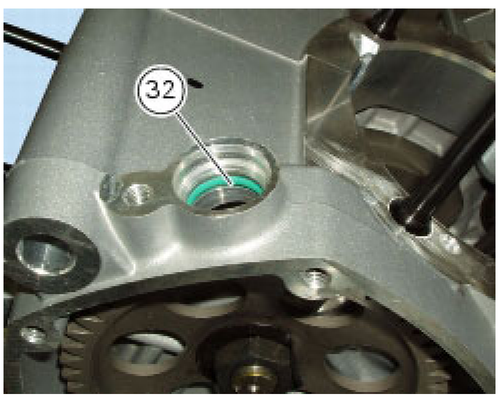
Refit the mesh filter (18) with its seal as described in Section D 4, Changing the engine oil and filter cartridge.



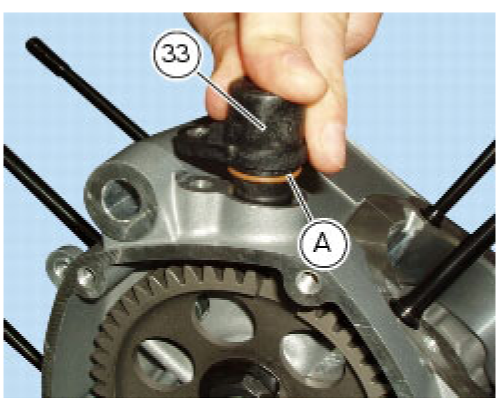
Now fit the stud bolts (12) in the crankcase halves, applying threadlocker on the threads and tightening to the specified torque (Sect. C 3, Engine torque settings). Use a tool similar to the one shown.



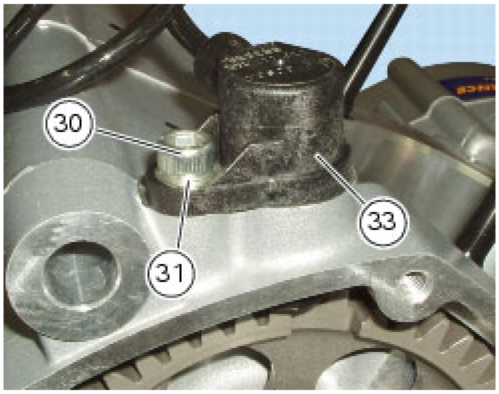
Check that the O-ring (32) is installed in the crankcase.



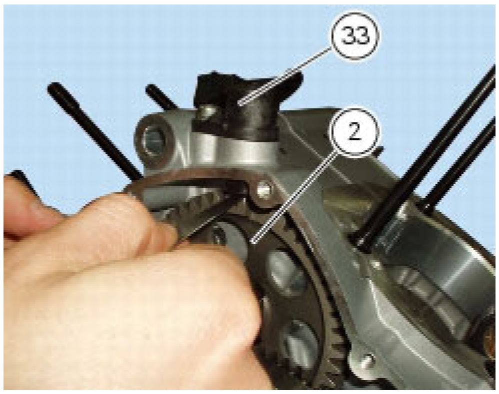
Grease the pick-up sensor (33) and ensure the O-ring (A) is in place.



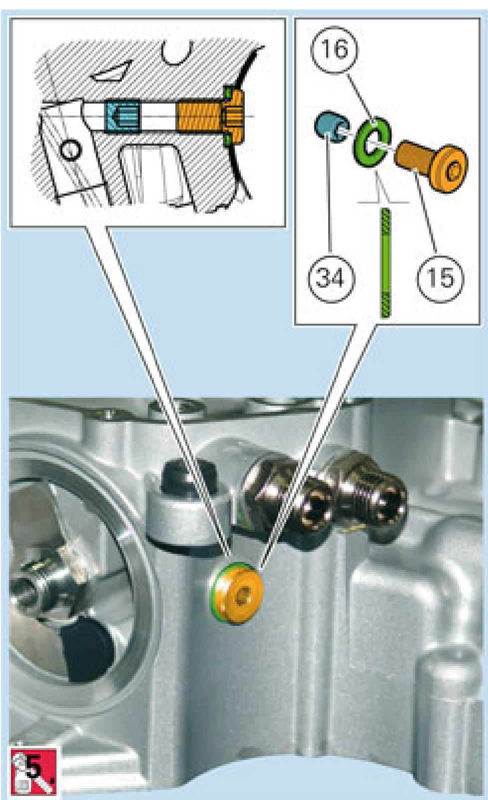
Fit the engine pickup (33) in its seat in the crankcase half.
Insert the screw (30) with the washer (31) and tighten to the specified torque (Sect. C 3, Engine torque settings).



Use a feeler gauge to check the clearance between the engine sensor (33) and the timing gear (2). The value must be between **0.6** and **0.8** mm.



If removed, apply prescribed threadlocker on the dowel (34), tighten the dowel to the specified torque (Sect. C 3, Engine torque settings), insert the gasket (16) on the service plug (15): the seal must be oriented so that the square edge faces the clutch-side crankcase half. Apply prescribed threadlocker on the plug thread (15), insert it in the crankcase half and tighten to the specified torque (Sect. C 3, Engine torque settings).



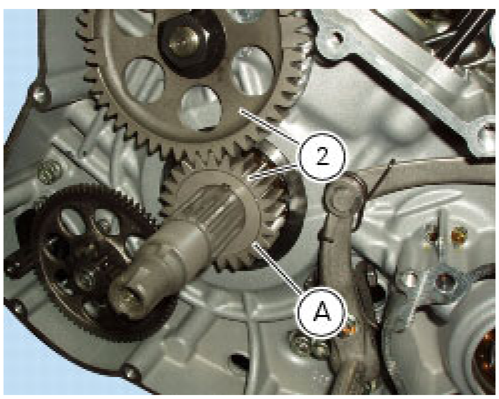
Operation	Section reference
Refit the starter motor	P 3, Refitting the starter motor

Refit the alternator-side crankcase cover and alternator assembly	N 8, Refitting the generator cover
Refit the cylinder barrel/piston assemblies	N 5, Refitting the cylinder/piston assembly
Refit the complete cylinder head unit	N 4.4, Refitting the cylinder head assembly
Refit the camshafts	N 4.3, Refitting the camshafts
Refit the timing belt pulleys and the timing belts	N 4.2, Refitting the timing system assembly
Refit the lubrication system	N 2.1, Refitting the oil pump
Refit the engine to the frame	N 1, Refitting the engine

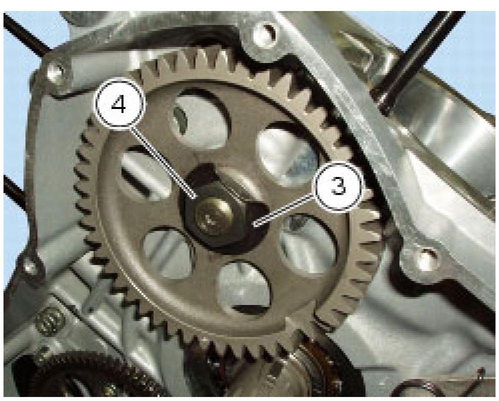
Removal of the timing gears

Operation	Section reference
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Remove the gearchange control	F 5, Removal of the gearchange control
Remove the clutch slave cylinder	F 2, Removal of the clutch slave cylinder
Remove the front sprocket cover	G 8, Removing of the front sprocket
Remove the generator cover and flywheel-generator assembly	N 8, Removal of the generator cover

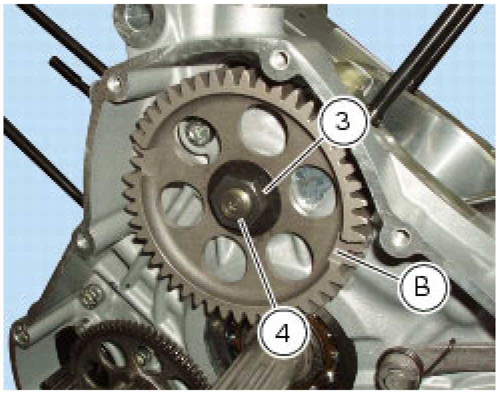
Slide out driving gear (A) of timing gear pair (2).



Relieve the staking on the lock washer (3) of the nut (4) that locks timing gear in place.
Restrain the driven timing gear by inserting a pin in one of the holes, and unscrew the nut (4).



Remove the nut (4), the washer (3) and the timing driven gear (B).

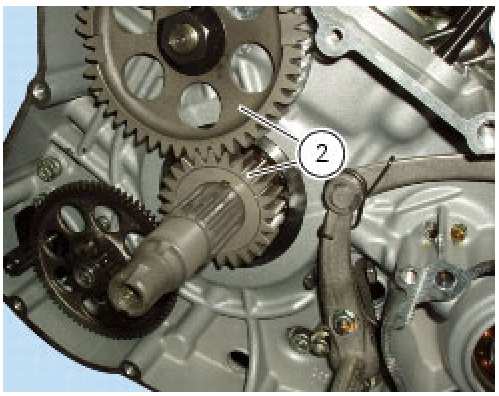


Refitting the timing gears

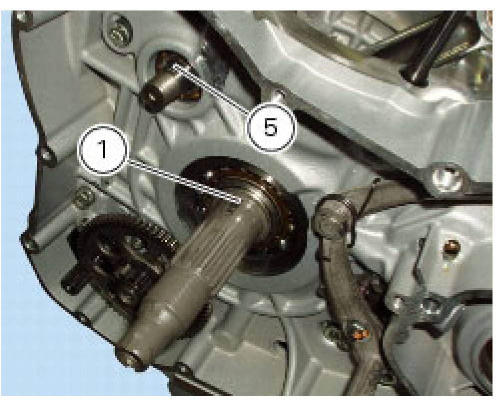
Before refitting, check the wear on the timing gear pair (2) and renew if necessary.

Important

The timing gears (2) must always be renewed as a pair.



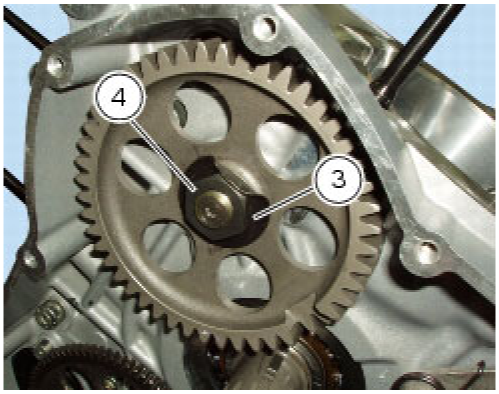
Check that the key (1) is correctly fitted on the timing belt driveshaft and that key (5) is correctly fitted on the crankshaft.



Refitting is the reverse of removal.

Important

On completion of the refitting operations, check that tab washer (3) is staked against nut (4) in such a way as to prevent the nut from working loose.

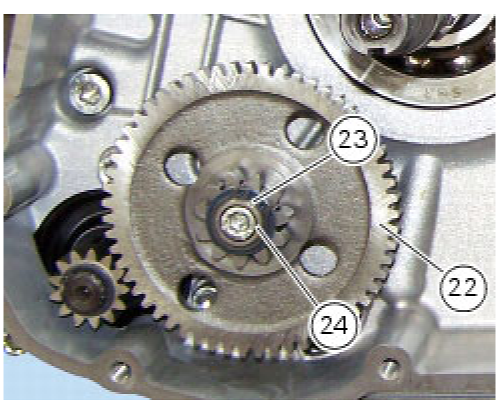


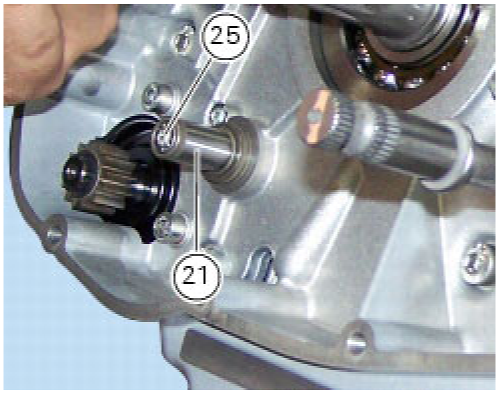
Operation	Section reference
Refit generator cover and the flywheel/generator assembly	N 8, Refitting the generator cover
Refit the sprocket cover	G 8, Refitting the front sprocket
Refit the clutch slave cylinder	F 2, Refitting the clutch slave cylinder
Refit the gearchange control	F 5, Refitting the gearchange mechanism
Fill the engine with oil	D 4, Changing the engine oil and filter cartridge

Removal of the starter motor idler gear

Operation	Section reference
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Remove the gearchange control	F 5, Removal of the gearchange control
Remove the clutch slave cylinder	F 2, Removal of the clutch slave cylinder
Remove the front sprocket cover	G 8, Removing of the front sprocket
Remove the generator cover and flywheel-generator assembly	N 8, Removal of the generator cover

Remove the circlip (24) and the washer (23).
Remove the starter idle gear (22).
Unscrew the screw (25) retaining idle gear shaft (21) and remove the shaft.



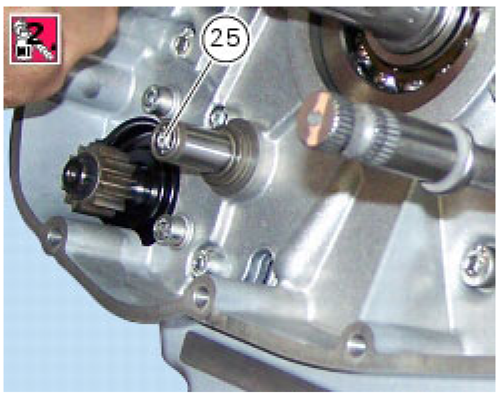


Refitting the starter motor gear

Refitting is the reverse of removal.

Note

Smear some threadlocker on screw (25) and tighten it to the specified torque (Sect. C 3, Engine torque settings).



Operation	Section reference
Refit generator cover and the flywheel/generator assembly	N 8, Refitting the generator cover
Refit the sprocket cover	G 8, Refitting the front sprocket
Refit the clutch slave cylinder	F 2, Refitting the clutch slave cylinder
Refit the gearchange control	F 5, Refitting the gearchange mechanism
Fill the engine with oil	D 4, Changing the engine oil and filter cartridge